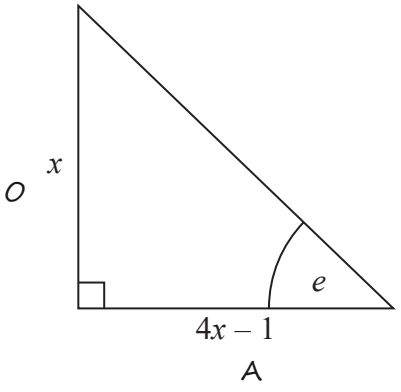
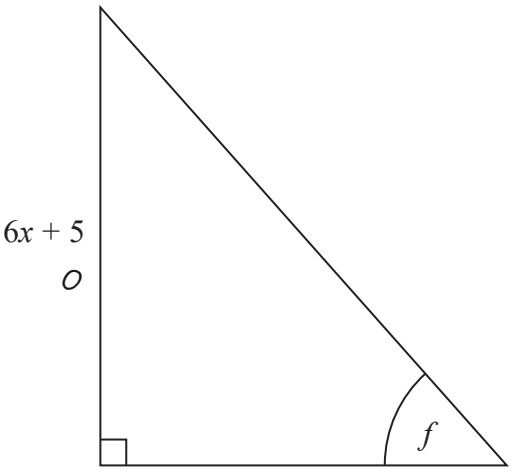
19 Here are two right-angled triangles.



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Given that

$$\tan e = \tan f$$

find the value of x.

You must show all your working.

tan = opposite ÷ adjacent, so

$$\frac{x}{4x - 1} = \frac{6x + 5}{12x + 31}$$

1 mark
Use the two tangent ratios

$x(12x + 31) = (6x + 5)(4x - 1)$

1 mark
Multiply by both denominators

	$12x$	31
x	$12x^2$	$31x$

	$4x$	-1
$6x$	$24x^2$	$-6x$
5	$20x$	-5

$12x^2 + 31x$

$= 24x^2 + 14x - 5$

$0 = 12x^2 - 17x - 5$

1 mark
Collect into a quadratic

	$4x$	1
$3x$	$12x^2$	$3x$
-5	$-20x$	-5

$12 \times (-5) = -60$; use -20 and $+3$.

$(4x + 1)(3x - 5) = 0$

1 mark
Factorise using the grid

$x = -\frac{1}{4}$ or $x = 5 \div 3$.

1 mark
Keep the positive solution

Reject $-\frac{1}{4}$: x must be positive.

$\frac{5}{3}$

(Total for Question 19 is 5 marks)